

Jordan E. Martinez

Albuquerque | (505)508-9863 | jrdnmrtnz12@gmail.com | [LinkedIn](#) | [GitHub](#)

SKILLS

- Programming & Scripting: Python, C/C++, Java, SQL, Bash/Shell Scripting
- Software & AI/ML Concepts: Data Structures & Algorithms, Object Oriented Programming (OOP), Genetic Algorithms, Satisfaction Problems (CSP), Search Algorithms (Minimax/Alpha-Beta), API Design
- Automation & DevOps/Tools: Infrastructure Automation, Source-of-Truth Synchronization, Git/GitHub, ServiceNow, Linux/Unix, RabbitMQ, CI/CD Fundamentals
- Networking & Protocols: Enterprise Networking, Out-of-Band Management, Storm Control, Routing & Switching, TCP/IP

EXPERIENCE

Sandia National Laboratory — Year-Round Technical Intern - Enterprise Networking
Albuquerque, NM

October 2023 - Present

- Network Automation: Automated source-of-truth processes across three enterprise networks and automated out-of-band network connectivity workflows to reduce manual intervention and configuration drift.
- Network Reliability & Security: Implemented storm control protocols across enterprise switches to prevent traffic overload and enhance network resilience.
- Ticketing Automation: Built a ServiceNow-based port activation workflow and connected it to a RabbitMQ message broker, enabling ticket results to be consumed by a Python automation script for automated network changes.

University of New Mexico — IT Technician - UNM Learn Management Systems
Albuquerque, NM

September 2021 - May 2023

- Defect Reproduction & Diagnostics: Replicated complex software issues in staging/test environments to analyze user workflows, isolate root causes, and resolve system anomalies within the UNM Learn LMS platform.
- Technical Support & Incident Management: Provided high-tier technical resolution for students and faculty, diagnosing software errors, access permissions, and application integration bugs.
- Root Cause Analysis: Collaborated with team members to develop and document software workarounds, technical fixes, and escalation paths to ensure minimal system downtime.
- User-Centric Troubleshooting: Empathized with end-user pain points to refine user journey workflows, offering actionable feedback to improve application usability and technical accessibility.

EDUCATION

University of New Mexico

Expected December 2026

- Degree: MS - Computer Science | GPA – 3.95

Completed December 2025

- Degree: BS - Computer Science, double minor in Mathematics and Psychology
- Honors: Cum Laude

ACADEMIC PROJECTS

1. Evolving NFL play-calling strategies with a genetic algorithm (Complex Adaptive Systems, Spring 2026)

- Built a football drive simulator in Python using 106,000+ plays of 2022–2024 NFL play-by-play data (nfl_data_py). It models down and distance, field position, and situational play-calling.
- Implemented a genetic algorithm (tournament selection, uniform crossover, Gaussian mutation, elitism) that evolved offensive strategies over 100 generations.
- Found and corrected a fitness-function exploit across five experiments. Adding a turnover penalty flipped the evolved strategy from deep-pass dominant to short-pass dominant.
- Validated the results across 10 random seeds (50 GA runs). This showed a rugged fitness landscape with bimodal convergence.

2. Security analysis of AI-generated authentication code (Cyber Security, Spring 2026)

- Evaluated 30 Flask authentication code samples from ChatGPT, Gemini, and Claude Code under naive and security-aware prompts.
- Combined Bandit static analysis with manual review mapped to OWASP Top 10 and CWE. Manual review found 114 issues versus Bandit's 21, including six of seven Critical findings that Bandit missed.
- Identified cross-file security inconsistencies, such as correct password hashing in one file and plaintext comparison in another, that static tools can't detect.

3. Multi-elevator control and simulation system (Software Engineering, Spring 2025)

- Developed components of a multi-elevator control system in Java, integrating elevator cabins, buttons, door assemblies, operating modes, and notification systems.
- Implemented controller functionality and Command Center UI features for monitoring and managing elevator state and operating modes.
- Collaborated with a multi-team development group to integrate independently developed subsystems, debug system-level issues, and conduct code reviews.
- Contributed to requirements and software architecture documentation, including RDD and SAD diagrams describing component responsibilities and interactions.

4. Parallel matrix multiplication optimization (Parallel Processing, Fall 2025)

- Implemented and benchmarked baseline, loop-unrolled, and cache-blocked matrix multiplication algorithms in C/C++
- Reduced execution time for 1000×1000 matrix multiplication from approximately 0.55 seconds to 0.20 seconds, achieving a 2.75× speedup over the baseline.
- Analyzed memory-access patterns, cache locality, and loop-level optimizations to improve computational performance.
- Benchmarked implementations on an HPC environment to compare optimization techniques and quantify performance improvements.